

Data Flow Diagram

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Data Flow Diagram

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system, showing the flow of information between users, processes, external services, and data stores. In the **Emotion Detection & Learning Support Engine**, the DFD illustrates how the user enters text describing their emotions or learning challenges, which is then validated and pre-processed by the system. The processed text is analyzed using a fine-tuned **BERT emotion detection model** to identify the user's emotional state and confidence score. If the user requests additional assistance, the query is forwarded to the **Gemini AI API**, which provides intelligent guidance and learning support. Finally, the detected emotion, recommendations, and AI-generated response are displayed to the user.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded Shape	External Entity	A person or external AI service that sends or receives data from the system (e.g., User, Gemini API).
Rectangle with Numbered Header	Process	An activity that transforms incoming data into meaningful output (e.g., Input Handling, Emotion Detection, Learning Support).
Cylinder / Data Store	Data Store	A place where information is stored for later use (e.g., User Input History, Resource Repository, Trained BERT Model).
Labeled Arrow	Data Flow	The movement of data between external entities, processes, and data stores, labeled with the type of data being transferred.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

